

Figure 1: OpenCV platforms

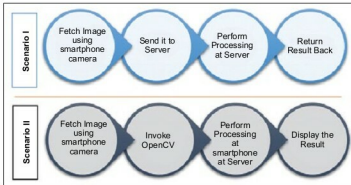


Figure 2: Computer vision scenarios

Computer vision at the smartphone level is carried out in two distinct ways, as shown in Figure 2.

One way of handling computer vision using low-end devices is to capture the images via the smartphone camera, send them over the network to a server, do the processing at the server end and return the results to the smartphone. This method includes a network overhead, which obviously delays the overall process. Another way of performing computer vision tasks is to use the smartphone's inherent processing power. With the improved hardware profiles of smartphones these days, it is possible to perform many of the computer vision tasks in the device itself.

OpenCV was ported to Android in 2010. The Android port of OpenCV provides full support for performing computer vision operations in smartphones. The OpenCV version 3.0 beta was released in November 2014. The stable OpenCV4Android version 2.4.10 can be downloaded from <https://sourceforge.net/projects/opencvlibrary/files/opencv-android/2.4.10/OpenCV-2.4.10-android-sdk.zip/download>.

OpenCV4Android is in the form of an SDK. The detailed documentation of the Java API is also included. Another interesting feature from the learner's perspective is that it includes a set of prebuilt sample *apk* files. These files can



Figure 3: The OpenCV4Android SDK

be loaded into any Android device and executed straight away. Thus, those learning about OpenCV4Android get an opportunity to see the potential use of this library, in action.

The structure of the OpenCV4Android SDK is shown in Figure 3.

The SDK folder has the OpenCV API and the Android libraries. The Java folder holds the OpenCV Java API, and the native folder has OpenCV C++ header files in addition to the

native Android libraries.

The Android OpenCV manager

To execute the OpenCV4Android applications in the device, the OpenCV manager needs to be loaded in the target device. This manager shares the libraries between various OpenCV applications running on the same device, and has been incorporated to reduce the overall memory requirements of multiple computer vision applications running in the device. It also offers the advantage of optimising for specific hardware. Due to the availability of the OpenCV manager, upgrading to newer versions becomes seamless across multiple applications. This manager can be freely downloaded from the Google Play store (<https://play.google.com/store/apps/details?id=org.opencv.engine>).

In the OpenCV4Android SDK, there is a folder called *apk* in which facilitates OpenCV library access through the OpenCV manager.

The flow of events for a typical OpenCV app is as listed below:

1. Download and install any app that relies on OpenCV, from the Google Play store.
2. If this is the first app of this kind in your device, then:
 - a. Download OpenCV manager from the Play store.
 - b. OpenCV manager gets started.
 - c. If required, the installation of an architecture-specific OpenCV library is suggested.

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